

Prompt Sensitivity Analysis of LLMs on Insurance Approval

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1 Introduction

Large language models (LLMs) are increasingly deployed in decision-support systems across domains such as healthcare, finance, and insurance. While these systems are often presented as neutral and objective, real-world reporting has raised concerns about automated decision pipelines influencing high-stakes outcomes. For example, recent investigations have highlighted the use of algorithmic systems in health insurance claim screening and alleged wrongful denials of care[1, 2]. These cases illustrate how automated systems can materially affect patient access to medical treatment and financial stability.

Beyond concerns about bias, another emerging issue is decision instability — whether model outputs change when the same decision rules are presented in different formats. Recent research has demonstrated that LLM performance can be highly sensitive to prompt phrasing and task framing, even when the underlying task remains unchanged[3]. Such sensitivity raises important questions about the reliability of LLM-based decision systems, particularly in structured and high-stakes contexts.

Motivated by real-world concerns surrounding automated insurance claim screening, this study investigates whether variations in the formatting of insurance decision rules influence LLM behavior. Rather than modifying the substantive meaning of policies, we hold policy intent constant while systematically varying presentation style. We examine whether formatting affects overall approval rates, the relationship between claimant attributes and model decisions, and the stability of outputs across experimental conditions.

2 Experimental Design

Our experiment evaluates how large language models (LLMs) behave when performing structured decision-making tasks under varying policy presentation formats. Specifically, we investigate whether differences in how insurance decision rules are written — rather than differences in their substantive content — influence model outputs.

2.1 Synthetic Claims Dataset

Due to the absence of publicly available datasets containing detailed individual-level health insurance claim decisions, we constructed a fully synthetic dataset consisting of 100 medical insurance claims. While all records are simulated, the structure of the dataset was informed by commonly used real-world insurance claim forms.

The synthetic dataset was designed to reflect typical insurance claim review inputs, including demographic attributes (e.g., age and gender), clinical indicators such as diagnosis severity, and administrative variables such as documentation completeness and claim amount (*see appendix A for full list of covariates*).

2.2 Model Selection

We evaluate three models from the **Gemma 3**, family of large language models developed by Google DeepMind: *Gemma 3 4B-it*, *Gemma 3 12B-it*, and *Gemma 3 27B-it*.^[4] These models were selected to examine whether decision behavior differs across model scales.

The B is an indicator of the parameter count of the model. A 4B model means that the model has 4 billion parameters. A parameter measures the number of variables the model uses to process data and response.

2.3 Policy Presentation Conditions

Each model evaluated the same set of synthetic claims under seven policy presentation conditions. Although the substantive decision criteria remained identical across conditions—including requirements related to medical necessity, coverage eligibility, and documentation completeness—the manner in which these criteria were presented to the models varied systematically.

The experimental design included a baseline condition (**placebo**) in which no explicit policy instructions were provided, allowing models to rely on their general reasoning capabilities. In the remaining conditions, policies were presented using different structural and linguistic formats intended to reflect realistic variations in administrative documentation. These included narrative **prose** descriptions, concise **bullet-point** summaries, and stepwise **decision-tree** formulations. Additional conditions explored the effects of framing and verbosity by emphasizing legal constraints such as non-discrimination (**bias-emphasis**), simplifying policies into **minimal** rule sets, or presenting them in highly detailed legalistic language (**verbose**).

By holding policy content constant while varying only presentation format, this design isolates the extent to which model decision behavior is sensitive to structural and rhetorical differences in policy communication.

2.4 Prompting and Execution Procedure

Each claim was provided to the model along with the corresponding policy instructions. Models were instructed to return a single categorical decision:

APPROVE, DENY, or NEED_MORE_INFO

Restricting outputs to a single label reduced variation due to explanatory text and simplified downstream statistical analysis.

Each model evaluated all claims under all policy conditions:

$$3 \text{ models} \times 7 \text{ policy conditions} \times 100 \text{ claims} = 2100 \text{ total decisions}$$

All experiments were conducted via the *Gemini API*[5] using a consistent prompt template. The model was asked to analyze all 100 claims at once in a single session, repeating this process 7 times for each of the different prompts.

3 Analysis

We begin with exploratory analysis to assess whether model decisions vary systematically across policy formats, model sizes, and claimant characteristics. Across all experiments, the models produced 2,100 total decisions (3 models $\times$ 7 policy formats $\times$ 100 claims).

3.1 Decision Distribution Across Models and Policy Formats

Table 1: Decision counts by model

model	Approve	Deny	Need More Info
gemma-3-4b-it	478	106	116
gemma-3-12b-it	422	205	73
gemma-3-27b-it	591	56	53

Overall approval rates differed substantially across both policy format and model size. The largest model (Gemma 27B) exhibited the highest approval rate and the lowest denial rate, while the smallest model (Gemma 4B) showed more balanced decision distributions.

The 4B model displayed the strongest variation across prompt formats, particularly under the bullet-point and minimal policies. By contrast, the 12B model produced relatively similar decision distributions across prompts, while the 27B model showed consistently high approval rates across most conditions.

Table 2: Decision counts by policy (4B)

policy	Approve	Deny	Need More Info
policy00 placebo	80	12	8
policy01 prose	74	5	21
policy02 bullets	41	48	11
policy03 decision tree	84	8	8
policy04 bias emphasis	70	5	25
policy05 minimal	57	17	26
policy06 verbose	72	11	17

Table 3: Decision counts by policy (12B)

policy	Approve	Deny	Need More Info
policy00 placebo	63	28	9
policy01 prose	59	31	10
policy02 bullets	55	35	10
policy03 decision tree	58	33	9
policy04 bias emphasis	58	25	17
policy05 minimal	56	31	13
policy06 verbose	73	22	5

A chi-square test of independence between policy format and decision outcome revealed that the effect of policy formatting differed by model size. In particular, policy format significantly influenced decisions for the 4B model ($p < 10^{-22}$) and the 27B model ($p = 0.0048$), but not for the 12B model ($p = 0.187$). This suggests heterogeneous sensitivity to prompt structure across model scales.

3.2 Policy Effects on Approval Probability

To quantify the effect of policy format on approval likelihood, we fit a generalized linear model with a binomial response:

$$\text{logit } P(\text{APPROVE}) = \beta_0 + \beta_1 \text{policy_id}.$$

Table 6 reports odds ratios for each policy format relative to the placebo condition. The bullet-point format significantly reduced approval odds (OR = 0.51, $p < 0.001$), and the minimal format also reduced approval odds (OR = 0.68, $p = 0.031$). Other policy formats did not produce statistically significant differences from the placebo condition. These findings suggest that formatting alone—without changing substantive decision rules—can alter model decision behavior.

Table 4: Decision counts by policy (27B)

policy	Approve	Deny	Need More Info
policy00 placebo	82	12	6
policy01 prose	83	0	17
policy02 bullets	85	9	6
policy03 decision tree	88	6	6
policy04 bias emphasis	83	11	6
policy05 minimal	88	6	6
policy06 verbose	82	12	6

Table 5: Chi-square test: policy ID x decision (per model)

model	statistic	degrees of freedom	p-value
gemma-3-4b-it	131	12	3.95e-22
gemma-3-12b-it	16.1	12	1.87e-1
gemma-3-27b-it	28.4	12	4.76e-3

Table 6: Odds ratios for approval by policy format relative to the placebo condition

Policy Format	Odds Ratio	p-value	95% CI
policy01 prose	0.857	0.405	[0.596, 1.23]
policy02 bullets	0.507	<0.001	[0.357, 0.717]
policy03 decision tree	1.10	0.634	[0.753, 1.59]
policy04 bias emphasis	0.790	0.200	[0.551, 1.13]
policy05 minimal	0.677	0.031	[0.473, 0.964]
policy06 verbose	1.04	0.850	[0.715, 1.50]

3.3 Administrative and Clinical Predictors of Approval

To assess which claim characteristics most strongly influenced model decisions, we estimated generalized linear models including both administrative and clinical variables. Across specifications, administrative verification signals demonstrated the largest and most consistent associations with approval probability.

In particular, documentation completeness exhibited the strongest effect across models, with incomplete claims overwhelmingly resulting in denial outcomes. This pattern suggests that models treat procedural completeness as a prerequisite condition for evaluation rather than as one factor among many.

Claim amount also showed a statistically significant association with ap-

proval probability, with higher or less plausible costs linked to lower approval likelihood. In contrast, clinical severity displayed weaker and model-dependent effects. While some models showed sensitivity to severity categories, these associations were not consistent across model scales.

Overall, these results indicate that administrative features exert a stronger influence on model decision behavior than clinical characteristics within the experimental setting.

3.4 Adjusted Bias Analysis

Because the claims dataset was synthetically generated, it is important to clarify how demographic attributes were incorporated. Gender was assigned independently as a descriptive attribute and was not used to determine clinical severity, documentation status, or claim amount during the generation process. While diagnosis categories were designed to remain medically plausible, no systematic rules linking gender to claim difficulty or approval likelihood were imposed. Nevertheless, because synthetic generation may implicitly reflect patterns learned by the generating model, observed demographic associations must be interpreted cautiously.

Table 7: Chi-square test of gender and decision outcome by model

Model	Statistic	df	p-value
gemma-3-4b-it	10.6	2	0.00503
gemma-3-12b-it	1.32	2	0.517
gemma-3-27b-it	14.9	2	0.000571

We first examined whether model decisions differed by gender using unadjusted contingency tables. A pooled chi-square test indicated a statistically significant association between gender and decision outcome ($p = 0.0097$). Model-specific tests suggested that this raw association was present for the 4B and 27B models but not for the 12B model (Table 7).

However, unadjusted comparisons may reflect differences in clinical or administrative claim characteristics rather than demographic effects. To assess whether gender remained associated with approval probability after controlling for these factors, we estimated the following generalized linear model:

$$\text{logit } P(\text{APPROVE}) = \beta_0 + \beta_1 \text{gender} + \beta_2 \text{policy_id} + \beta_3 \text{age} + \beta_4 \text{severity} \\ + \beta_5 \text{preexisting} + \beta_6 \text{docs_complete} + \beta_7 \text{claim_amount}.$$

After adjustment, gender was not significantly associated with approval probability in the pooled analysis (OR = 0.91, $p = 0.41$), and model-specific estimates likewise showed no statistically significant gender effect (Table 8). This suggests that the raw gender association observed in contingency tables was largely explained by differences in claim characteristics rather than by a consistent demographic effect in model decision behavior.

Table 8: Adjusted odds ratios for approval: gender (Male vs baseline gender (Female))

Model	Odds Ratio	p-value	95% CI
Pooled	0.909	0.410	[0.724, 1.14]
4B	0.702	0.104	[0.458, 1.07]
12B	1.01	0.964	[0.712, 1.43]
27B	1.11	0.726	[0.623, 1.96]

To further examine whether prompt formatting altered demographic effects, we compared a main-effects model with a model including gender-by-policy interaction terms:

$$\text{logit } P(\text{APPROVE}) = \beta_0 + \text{gender} \times \text{policy} + \text{covariates}.$$

A likelihood-ratio test did not indicate significant interaction effects in the pooled analysis ($p = 0.88$), nor within any individual model (4B: $p = 0.852$, 12B: $p = 0.566$, 27B: $p = 0.994$). These results suggest that policy formatting influenced overall approval behavior without producing evidence of prompt-specific gender disparities.

Beyond demographic variables, administrative claim features exhibited the strongest associations with approval probability. In particular, documentation completeness showed a consistently large positive effect across specifications, indicating that models treated documentation as a prerequisite condition for evaluation. Claim amount also demonstrated a smaller but statistically significant association, with higher or less plausible costs linked to reduced approval likelihood.

Other clinical and administrative variables, including diagnosis severity and prior authorization status, showed weaker or model-specific effects.

Overall, these findings indicate that model decisions were driven primarily by procedural verification signals and cost-related information rather than by demographic attributes. However, because the dataset was synthetically generated, it remains possible that latent correlations introduced during generation influenced observed associations. Future work should validate these findings using real-world claims data or controlled synthetic generation procedures that explicitly enforce demographic independence.

3.5 Interaction Between Policy Format and Model Size

We next tested whether the impact of policy formatting differs by model size via a likelihood ratio test comparing:

$$\text{logit } P(\text{APPROVE}) = \beta_0 + \text{policy} + \text{model} + \text{covariates}$$

versus

$$\text{logit } P(\text{APPROVE}) = \beta_0 + \text{policy} \times \text{model} + \text{covariates}.$$

The likelihood-ratio test comparing a model with policy and model main effects to a model including policy-by-model interactions was highly significant ($p = 4.4 \times 10^{-6}$). This indicates that prompt formatting does not affect all models equally. In particular, the 4B model was highly sensitive to policy presentation, the 27B model exhibited smaller but still significant formatting effects, and the 12B model showed relatively little variation across policy formats.

3.6 Decision Stability Across Policy Formats

To evaluate the robustness of model decisions to variations in policy presentation, we defined a claim-level *flip rate* as the proportion of policy conditions under which a claim received a different approval decision relative to its modal decision across the seven formatting conditions.

Formally, for each claim i , let $d_{i1}, \dots, d_{i7}$ denote the binary decisions under the seven policy formats. Stability was defined as

$$\text{Stability}_i = \frac{\text{number of decisions equal to the modal decision}}{7},$$

and instability (flip rate) was defined as

$$\text{Instability}_i = 1 - \text{Stability}_i.$$

Model-level stability was then computed as the average stability across all claims.

As shown in Table 9, the Gemma-3-27B model exhibited the highest decision consistency (stability rate = 0.88), whereas the Gemma-3-4B model demonstrated moderate sensitivity to policy formatting (stability rate = 0.50). In contrast, the Gemma-3-12B model showed substantial instability (stability rate = 0.01), indicating that its decisions frequently varied across presentation formats.

Table 9: Stability Rate of AI Decisions Across 7 Prompt Variations

model	Stability Rate	Instability Rate
gemma-3-4b-it	0.50	0.50
gemma-3-12b-it	0.01	0.99
gemma-3-27b-it	0.88	0.12

These results indicate substantial variation in decision robustness across model scales. The largest model maintained relatively consistent outcomes despite formatting changes, whereas smaller and mid-scale models exhibited greater sensitivity to prompt structure.

To further examine determinants of instability, we estimated a logistic regression model predicting claim-level flipping. Documentation completeness was associated with reduced instability ($p = 0.044$), suggesting that clear administrative verification signals contribute to more consistent decision behavior. Model scale remained the strongest predictor of stability overall.

Taken together, these findings suggest that policy formatting can influence decision reliability, but that robustness depends not only on model capacity but also on the presence of strong procedural signals in the input.

3.7 Summary of Empirical Results

The results provide preliminary evidence that the presentation structure of policy rules can influence model-generated insurance decisions even when the substantive criteria remain constant. This finding suggests that decision outcomes produced by large language models are not solely determined by underlying policy logic, but are also shaped by the way those rules are communicated within prompts.

In addition, model stability varied substantially across model sizes. Larger models demonstrated greater consistency in applying decision criteria across formatting variations, while smaller and mid-scale models exhibited higher sensitivity to prompt structure. This pattern indicates that model capacity may contribute to robustness, although scale alone does not guarantee stable behavior.

Administrative verification signals, particularly documentation completeness, emerged as dominant predictors of approval decisions. This suggests that models may prioritize procedural completeness over nuanced clinical reasoning when evaluating structured claims.

Future work should extend the analysis to multinomial outcome modeling, broader fairness metrics across additional demographic dimensions, and robustness checks using alternative prompt formulations. Such extensions would help clarify whether the observed effects generalize across decision contexts and model architectures

4 Interpretation of Results

The primary objective of this study was to evaluate whether large language models produce stable and reliable insurance claim decisions when identical policy rules are presented in different formats. The findings indicate that model behavior is sensitive to presentation structure and varies substantially across model scales, raising important considerations for the use of LLMs in structured decision-support environments.

4.1 Decision Stability and Model Scale

A central result of the analysis concerns the degree of decision stability across policy presentation formats. The largest model (Gemma 27B) demonstrated substantially higher consistency in its approval outcomes compared to the smaller models. In contrast, the mid-scale model (12B) exhibited near-complete instability, while the smallest model (4B) showed moderate variabil-

ity. These results suggest that model capacity is associated with robustness to formatting variation, although the relationship is not strictly monotonic.

Importantly, the observed instability in intermediate-scale models highlights that increasing model size alone does not ensure reliable decision-making. Instead, empirical validation of model behavior remains necessary prior to deployment in high-stakes institutional contexts.

4.2 Impact of Policy Formatting on Decision Outcomes

Across models, differences in policy presentation were associated with measurable changes in approval probabilities, even though the substantive decision rules were held constant. Formats that emphasized concise or structured rule representation tended to produce lower approval rates in pooled analyses. This pattern suggests that structural features of prompts may influence how models prioritize or interpret decision criteria.

From a systems perspective, this finding implies that automated decision pipelines incorporating LLMs may yield materially different outcomes depending on how policy rules are encoded. Such sensitivity raises concerns regarding procedural consistency, reproducibility, and fairness in operational settings such as insurance adjudication.

4.3 Role of Administrative and Clinical Features

The analysis results indicate that administrative verification signals play a dominant role in model decision behavior (Section 3.3). Documentation completeness emerged as the strongest predictor of approval probability across specifications, suggesting that models treat procedural completeness as a prerequisite for evaluation.

Similarly, claim amount demonstrated a consistent but smaller effect, with higher or less plausible costs associated with reduced approval likelihood. In contrast, clinical variables such as severity exhibited model-dependent associations, indicating variability in how different model scales operationalize clinical reasoning.

These findings suggest that LLM-based decision processes may rely heavily on procedural validation cues, potentially at the expense of substantive clinical considerations.

4.4 Demographic Fairness Considerations

Initial contingency analyses suggested an apparent association between gender and decision outcomes for certain models. However, these associations did not persist after adjusting for clinical and administrative covariates. This indicates that observed demographic disparities in raw decision distributions were largely attributable to differences in claim characteristics rather than to systematic demographic bias in approval behavior.

Furthermore, interaction analyses provided no evidence that policy formatting introduced prompt-specific gender disparities. Within the scope of this experimental design, formatting sensitivity appears to affect overall decision patterns rather than systematically amplifying demographic inequities.

4.5 Implications for AI-Assisted Decision Systems

Taken together, the findings highlight that LLM-based decision systems may be sensitive to non-substantive variations in rule presentation, potentially producing inconsistent outcomes in applied settings. While larger models exhibited greater stability, robustness cannot be assumed solely on the basis of scale. Instead, structured validation and standardization of prompting protocols are necessary to ensure reliable deployment.

The prominence of administrative completeness signals in shaping decisions further suggests that models may operationalize procedural criteria more strongly than clinical nuance. This imbalance may have important implications for how automated decision-support tools are integrated into institutional workflows, particularly in domains where decisions directly affect access to care and financial outcomes.

A Covariates

Each claim in the dataset includes the following covariates:

- Patient name
- Age
- Gender (sex at birth)
- Relationship to insured (e.g., self or dependent)
- Diagnosis or injury type
- Service type (e.g., emergency visit, outpatient care)
- Severity level (low, moderate, high)
- Preexisting condition status
- In-network provider status
- Prior authorization requirement and completion
- Documentation completeness
- Claim amount (USD)

These variables were selected to approximate factors commonly considered during insurance claim review processes.

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